

Epistemic Decay Rates in Iterated Knowledge Structures: A Category-Theoretic Perspective with Applications to Cybersecurity

Pearl Bipin Pulickal

Electronics Engineering Graduate, National Institute of Technology Goa

pearlbipin@gmail.com

29 January 2026

Abstract

Iterated statements of knowledge—constructions of the form “an agent knows that another agent knows”—arise naturally in modal logic, distributed systems, and game-theoretic security. While such constructions are formally iterable within standard logical frameworks, their semantic clarity and operational usefulness often degrade rapidly with increasing depth. In this paper, we investigate this phenomenon through a category-theoretic lens, introducing the notion of an *epistemic decay rate* (DK rate) to characterize the loss of coherence in higher-order knowledge structures. By modeling epistemic states as objects in a category of pointed coalgebras, we demonstrate that for finite state spaces, the filtration of epistemic distinctions must stabilize, leading to inevitable saturation. Focusing on four canonical configurations involving observers and objects, we show that beyond a small, structurally determined number of iterations, additional layers cease to contribute actionable information. Although our analysis is mathematical rather than psychological, it yields a rigorous justification for the “bounded horizons” observed in adversarial settings such as cybersecurity.

Keywords: category theory, epistemic logic, decay of knowledge, iterated knowledge, cybersecurity, adversarial reasoning, pointed coalgebras

1 Introduction

Reasoning about knowledge of knowledge—statements of the form “agent A knows that agent B knows X ”—is central to logic, distributed computation, and security analysis. Such iterated epistemic constructions appear naturally in adversarial environments, where agents continuously

adapt their strategies based on beliefs about the beliefs of others. In principle, formal systems such as standard modal logic (S5) allow these constructions to be iterated indefinitely. In practice, however, higher-order knowledge often exhibits a rapid loss of clarity, relevance, or operational meaning.

This paper is motivated by a simple but robust observation: while first- and second-order knowledge claims are frequently meaningful, deeper iterations tend to destabilize quickly. We refer to this phenomenon as the *epistemic decay rate* (DK rate). Rather than treating epistemic depth as an unbounded resource, we pose a structural question: *how many layers of iterated knowledge can a given configuration meaningfully sustain before collapsing into redundancy?*

Our approach is deliberately structural. We do not appeal to cognitive limitations, bounded rationality, or the psychological constraints of human agents. Instead, we analyze how instability arises from the relationships between observers and the objects of knowledge themselves. Category theory provides the natural language for this analysis, allowing us to treat knowledge updates as morphisms and iterated knowing as the recursive composition of endofunctors.

A central contribution of this work is a negative result: beyond a small number of iterations, higher-order knowledge ceases to add actionable information. This result does not depend on the intelligence or computational power of the agents involved, but rather on the internal coherence of the epistemic structure being iterated. We prove this not as a psychological observation, but as a consequence of stabilization in finite categories.

The remainder of this paper develops this idea in a controlled and incremental manner. We first introduce a minimal conceptual framework and four representative cases of epistemic iteration. We then formalize these notions using the category of pointed Kripke models, distinguishing between formal iterability and operational meaningfulness. Finally, we apply these insights to cybersecurity, where understanding the limits of epistemic depth can inform both attack and defense strategies.

2 Related Work

Our work sits at the intersection of epistemic logic, dynamic epistemic frameworks, and compositional (category-theoretic) modeling, with applications to security. We briefly survey the most relevant strands of literature.

Foundations of epistemic logic and common knowledge. The formal study of knowledge in multi-agent systems has a long and well-established foundation. Classic texts frame knowledge using modal operators and Kripke semantics and analyze constructs such as common knowledge; see Fagin et al. for a comprehensive treatment and Aumann for the seminal result on common knowledge and agreement [1, 2].

Dynamic epistemic logic and public announcements. Beyond static knowledge, a large body of work studies how knowledge changes with announcements and actions. The dynamic epistemic logic (DEL) literature develops tools for reasoning about information change (public announcements, private communication, and action models), providing a formal backbone for iterated epistemic reasoning and its transformations [3, 4, 5].

Epistemic approaches to security protocols. Epistemic-style logics have long been used to analyze security and authentication protocols: BAN logic inaugurated formal belief-based analyses of authentication protocols, and more recent efforts recast security protocol verification in DEL-style frameworks or specialized epistemic systems for protocols [6, 7, 8].

Category-theoretic and compositional perspectives. There is growing interest in recasting modal and dynamic epistemic semantics categorically, which both clarifies structural properties and enables compositional reasoning. Kishida provides a categorical perspective on DEL; other work has explored category-theoretic approaches to system engineering and secure design, arguing that compositional methods help manage complexity in cyber-physical and distributed systems [9, 10, 11]. These lines motivate our choice to phrase DK rates in categorical language.

Security economics and epistemic considerations. Complementary literature emphasizes decision-theoretic and economic aspects of security. The Gordon–Loeb model and its successors formalize why over-investment in deep but brittle reasoning can be suboptimal, offering a different angle on why bounded epistemic horizons may be rational in practice [12].

Summary. Our contribution draws together these strands: we adopt the formal richness of epistemic logic, add a lightweight categorical modeling layer to make the instability of iterated knowledge explicit, and point to implications for security design.

3 Conceptual Framework and Canonical Epistemic Configurations

In this section, we introduce the minimal conceptual vocabulary required to study the degradation of iterated knowledge. Our goal is not to construct a comprehensive epistemic logic, but to isolate the structural features that govern the stability or instability of higher-order knowledge claims.

⁰This structural bound provides a rigorous mechanism for the *Gordon-Loeb* result. Gordon and Loeb argue that optimal security investment does not increase linearly with vulnerability. In our terms, investment in epistemic depth n yields diminishing returns of information $I(n)$. Once $n > N$, the marginal information gain $\Delta I = 0$, making any cost $C(n)$ for $n > N$ a purely irrational “over-investment.” The “Decay Rate” is thus the derivative of the information utility curve.

3.1 Observers, Objects, and Epistemic Layers

We begin by fixing three basic notions.

Observers. An *observer* is any agent capable of holding and updating statements of the form “knows X .” No assumptions are made regarding rationality or cognitive capacity. Observers are treated as abstract loci of knowledge.

Objects of knowledge. An *object* is anything that can be known by an observer. Objects fall into two broad classes:

- *Neutral objects*, which are not intrinsically tied to any observer (e.g., mathematical facts, public data).
- *Non-neutral objects*, which are relationally dependent on one or more observers (e.g., another agent’s secret, an agent’s internal state).

Epistemic layers. An *epistemic layer* is formed by iterating the “knows” relation. A base-layer statement has the form “ A knows X .” Higher layers are constructed recursively, yielding expressions such as “ A knows that B knows X .”

3.2 Epistemic Decay Rate

While epistemic layers are formally iterable, their semantic coherence is not guaranteed to persist under iteration. We introduce the term *epistemic decay rate* (DK rate) to describe how rapidly meaningful structure degrades as epistemic depth increases.

Informally, the DK rate measures how many successive layers of “knowing that” can be constructed before the resulting expression becomes unstable, ambiguous, or operationally void. A high DK rate corresponds to rapid collapse under iteration, whereas a low DK rate indicates relative stability. Crucially, the DK rate is a *structural property*. It depends on the number of observers involved and on whether the object of knowledge remains neutral across layers.

4 A Categorical Model of Epistemic Iteration

In this section, we introduce a rigorous categorical framework to formalize epistemic iteration. To satisfy the need for mathematical grounding, we explicitly define the category $\mathcal{E}$ in terms of coalgebraic semantics, which is the standard categorical approach to modal logic.

4.1 Defining the Category $\mathcal{E}$

We define the category of epistemic states, denoted $\mathcal{E}$, as the category of ****Pointed Kripke Models**** (or equivalently, **Pointed Coalgebras**).

Definition (Category $\mathcal{E}$). Let $\mathcal{E}$ be the category where:

- An **object** is a triple (W, R, w_0) , where W is a set of possible worlds (states), $R \subseteq W \times W$ is an accessibility relation (representing knowledge/uncertainty), and $w_0 \in W$ is the designated “actual” world.
- A **morphism** $f : (W, R, w_0) \rightarrow (W', R', w'_0)$ is a bounded morphism (p-morphism) satisfying:
 1. $f(w_0) = w'_0$ (preserves the actual world);
 2. If uRv , then $f(u)R'f(v)$ (forward condition);
 3. If $f(u)R'v'$, there exists $v \in W$ such that uRv and $f(v) = v'$ (backward condition).

Intuitively, a morphism in $\mathcal{E}$ represents a simulation or abstraction of one epistemic state by another. This grounding ensures that our discussion of “iteration” has a concrete semantic model.

4.2 Knowledge as an Endofunctor

We model the operation “knows that” as an endofunctor $K : \mathcal{E} \rightarrow \mathcal{E}$. For an object $X = (W, R, w_0)$, the object $K(X)$ represents the epistemic state restricted to the reachable knowledge of the agent. In a multi-agent setting, we may view K_i as a functor focusing on the relation R_i .

Iterated knowledge is represented by repeated application:

$$X \xrightarrow{\eta} K(X) \xrightarrow{K(\eta)} K^2(X) \rightarrow \dots$$

where the maps represent the embedding of deeper epistemic layers.

4.3 Neutral and Non-Neutral Objects

We distinguish between two classes of objects within $\mathcal{E}$. An object O is *neutral* if it is isomorphic to a constant functor or a rigid object invariant under epistemic reflection (e.g., a frame with a single reflexive point). An object is *non-neutral* if it contains structure that varies under R , implying that the “knowledge of” the object is structurally distinct from the object itself.

4.4 Epistemic Decay as Failure of Stabilization

In categorical dynamics, iteration of an endofunctor often stabilizes via fixed points (limits or colimits). Epistemic decay corresponds to the *saturation* of this process. Specifically, rapid decay occurs when the sequence of refined partitions of the state space W stabilizes quickly.

We interpret this collapse as a loss of epistemic distinction. Although $K^n(X)$ may exist formally for all n , if $K^n(X) \cong K^{n+1}(X)$ (or if the morphisms become isomorphisms), the additional layer provides no new information.

4.5 The Four Canonical Cases

The four canonical cases correspond to distinct categorical behaviors:

- **Case 1 (Two observers, non-neutral):** Iteration induces a cycle between two relations R_A and R_B , rapidly saturating the filtration of W .
- **Case 2 (Two observers, neutral):** Iteration factors through a constant object, preserving stability but eventually yielding no new subsets of W .
- **Case 3 (One observer, initially neutral):** Neutrality is lost after one step (O becomes $K(O)$), leading to immediate self-referential collapse.
- **Case 4 (One observer, self-knowledge):** Iteration is purely endomorphic (R_A acting on R_A).

4.6 A Structural Bound on Epistemic Depth

We now formalize the main claim of the paper. This result explains why infinite iteration is mathematically possible but operationally finite.

Proposition (Epistemic Saturation). *Let $X = (W, R, w_0)$ be an object in $\mathcal{E}$ where W is a finite set. There exists a finite integer N such that for all $n \geq N$, the epistemic state $K^n(X)$ provides no new operational distinctions compared to $K^N(X)$.*

Proof (via Terminal Coalgebra Sequence). We model the epistemic system as a coalgebra for the powerset functor $F(X) = \mathcal{P}(At) \times \mathcal{P}(X)^k$ (for k agents and atomic facts At). The evolution of knowledge corresponds to the terminal sequence of this functor:

$$1 \xleftarrow{!} F(1) \xleftarrow{F(!)} F^2(1) \dots \leftarrow F^n(1) \dots$$

The object $F^n(1)$ represents the distinctions visible up to modal depth n . The canonical map $c_n : W \rightarrow F^n(1)$ assigns to each state its behavioral description at depth n .

Since the state space W is finite with cardinality $|W|$, the number of possible partitions of W is finite (specifically, bounded by the Bell number $B_{|W|}$). The sequence of equivalence relations $\sim_n$ induced by c_n is monotonically refining: if $u \sim_{n+1} v$, then $u \sim_n v$.

Strict refinement implies $|\mathcal{P}(W)/\sim_{n+1}| > |\mathcal{P}(W)/\sim_n|$. Since the maximum number of equivalence classes is $|W|$, the sequence must stabilize. That is, there exists an $N \leq |W|$ such that the map $c_{N+1} \rightarrow c_N$ is an isomorphism on the image.

Therefore, for all $n > N$, no new epistemic subsets are generated. The system has reached the final coalgebra for the restricted finite domain. $\square$

4.7 Quantifying the Decay Rate

The critique of “level- k ” reasoning often lacks a metric. Based on the stabilization proof, we propose the following upper bound for the useful depth N in a connected epistemic graph.

Let $G = (W, R)$ be the Kripke frame. We define the **Effective Epistemic Horizon** (or Decay Bound N) by the graph diameter and the branching factor. For a frame with worst-case branching factor b (ambiguity) and state size $|W|$:

$$N \approx \frac{\log |W|}{\log b} \quad (1)$$

Where:

- $|W|$ is the total number of possible world-states (e.g., “Attack” vs “Safe”).
- b is the measure of epistemic uncertainty (how many worlds are accessible from a current state).

If $b = 1$ (perfect information), $N = 0$ (immediate knowledge). As b increases (high uncertainty), N decreases. Counter-intuitively, high uncertainty makes deep reasoning *less* useful because the “cone” of possibilities expands to cover W rapidly, rendering distinct worlds indistinguishable at depth.

We define the **Decay Rate** λ as the inverse of the horizon:

$$\lambda \propto \frac{\log b}{\log |W|}$$

This formalizes why “Shared Secret” scenarios (Case 1, high b) decay faster than “Public Protocols” (Case 2, low b).

5 A Categorical Elaboration of the Canonical Configurations

We now revisit the canonical configurations using explicit diagrams in $\mathcal{E}$. These diagrams illustrate how composition leads to cycles or stabilization.

Case 1: Two Observers with a Non-Neutral Object (Fast Decay)

Let X and Y denote two observers (or their local state spaces). The arrows represent knowledge morphisms (accessibility).

$$X \xrightarrow{k_{XY}} Y \xrightarrow{k_{YX}} X$$

Under iteration, composition produces a cycle:

$$X \xrightarrow{k_{XY}} Y \xrightarrow{k_{YX}} X \xrightarrow{k_{XY}} Y$$

The diagram collapses into a cycle. Because the object is non-neutral (entangled), the observers must constantly re-evaluate each other. However, per the Proposition above, the information gained from this cycling saturates quickly.

Case 2: Two Observers with a Neutral Object (Slow Decay)

Let O be a neutral object (e.g., a constant frame).

$$X \xrightarrow{k_X} O \xleftarrow{k_Y} Y$$

Higher-order knowledge factors through the neutral object:

$$X \xrightarrow{k_X} O \xrightarrow{id} O \xrightarrow{id} O$$

Here, iteration stabilizes on the identity morphism of O . This factorization delays collapse compared to Case 1, as the internal states of X and Y are not recursively queried.

Case 3: Single Observer with an Initially Neutral Object (Very Fast Decay)

At layer 1, X knows O (neutral). At layer 2, X knows [that X knows O]. The object has effectively changed from O to $K_X(O)$.

$$X \xrightarrow{k} O \implies X \xrightarrow{k'} X$$

The loss of neutrality forces a transition to self-maps (endomorphisms) immediately.

Case 4: One Observer with a Non-Neutral Object (Immediate Collapse)

Here, the object is the observer itself from the start.

$$X \curvearrowright_k$$

All iterations are powers of the same endomorphism $k : X \rightarrow X$. If k is idempotent or nilpotent (common in logic), decay is instantaneous.

6 Iteration, Finiteness, and Structural Limits

The structural elaboration above highlights the tension between formal syntax and semantic reality.

6.1 Formal Iterability versus Operational Meaningfulness

We distinguish between:

- **Formal Iterability:** The syntactic ability to construct $K^n(X)$ for any n .
- **Operational Meaningfulness:** The condition that $K^n(X) \not\cong K^{n-1}(X)$ in a way that affects decision-making.

As proven in Section 4.6, operational meaningfulness is strictly bounded for finite systems. The "infinite" chains of knowledge posited by game theory often map to a single stable state in the underlying category $\mathcal{E}$ after a small index N .

6.2 A Negative Result on Higher-Order Knowledge

This leads to our central negative result:

Negative Result. *In any finite epistemic system, there is a hard limit on the depth of useful adversarial reasoning. Beyond this limit, "thinking deeper" is indistinguishable from "thinking the same."*

7 Application to Cybersecurity

Cybersecurity is an adversarial domain where agents often attempt to out-think opponents. Our framework suggests that such "level- k reasoning" has diminishing returns.

7.1 Adversarial Knowledge Layers

A defender might reason: "The attacker knows I have a firewall ($K_A K_D F$)."

The attacker might reason: "The defender knows that I know they have a firewall ($K_D K_A K_D F$)."

Our model predicts that for $n > 3$, these statements likely map to the same partition of the state space. They are distinct sentences, but they imply the exact same set of possible worlds.

7.2 Mapping Cybersecurity Scenarios to Canonical Cases

1. **Shared Secrets (Case 1):** High entanglement. Fast decay. Mind games about passwords or keys saturate immediately.
2. **Public Protocols (Case 2):** Neutral objects. Slower decay. One can reason deeply about the abstract protocol, but not about the specific agents running it.
3. **Self-Assessment (Case 4):** Defenders reasoning about their own security posture ($K_D K_D S$) gain little from iteration. A system scan provides ground truth; iterating beliefs about the scan does not.

Canonical Case	Cybersecurity Analog	Epistemic Behavior
Case 1 (2 Observers, Non-Neutral)	Shared Secrets (Keys, Passwords, Insider Threats)	High Entanglement. Immediate saturation. $K_A K_B S$ collapses to "secret is shared."
Case 2 (2 Observers, Neutral)	Public Protocols (TCP/IP, TLS Handshake Logic)	Stability. Deep reasoning is possible about the <i>protocol</i> , but not the <i>agent</i> .
Case 3 (1 Observer, Unstable)	Phishing / Social Eng. (Target's belief about sender's identity)	Fast Decay. Neutrality is lost once the user engages; belief becomes self-referential.
Case 4 (1 Observer, Self-Object)	Self-Assessment (Compliance Audits, Vulnerability Scans)	Immediate Collapse. K^n adds no value over K^1 (the scan result).

Table 1: Mapping of categorical epistemic configurations to cybersecurity domains. The stability of the object determines the strategic horizon.

7.3 Case Study: Epistemic Collapse in the Needham-Schroeder Protocol

To illustrate the operational reality of the decay bound N , we analyze the man-in-the-middle attack on the Needham-Schroeder Public-Key Protocol (NSPK).

In the classic attack discovered by Lowe, an attacker M impersonates agent A to agent B . The protocol failure is fundamentally epistemic: B believes he is talking to A ($K_B(\text{Partner} = A)$), but fails to verify that A knows she is talking to B ($K_B K_A(\text{Partner} = B)$).

We can model the "fix" (the Needham-Schroeder-Lowe protocol) as forcing the epistemic structure to stabilize at depth $N = 2$.

- **Pre-Fix (Unstable):** The message content does not include the identity of the responder. The epistemic state w (who is talking to whom) remains ambiguous in the higher-order beliefs. The chain $K_B K_A(\dots)$ cycles indefinitely between possible worlds where A is

talking to M or A is talking to B . The partition never stabilizes ($N \rightarrow \infty$), creating a vulnerability window.

- **Post-Fix (Stable):** By including the responder’s identity B in the second message ($A \leftarrow \{N_A, N_B, B\}_{PK_A}$), the protocol forces the object of knowledge to become *rigid*. The message creates a public fact:

$$K_A(\text{Partner} = B) \implies K_B K_A(\text{Partner} = B)$$

The inclusion of the identity forces the filtration of worlds to separate the “Attack” world from the “Honest” world immediately at step 2. The system achieves **Epistemic Saturation** at $N = 2$.

This confirms our structural hypothesis: robust security protocols are machines designed to force low- N saturation, converting deep uncertainty (Case 1) into stable facts (Case 2).

7.4 Strategic Horizons and DK Bounds

The *Epistemic Decay Bound* acts as a strategic horizon. Resources spent computing reasoning strategies beyond this bound are wasted. Effective defense systems should optimize for robustness at low epistemic depth (e.g., zero-trust architectures which assume breach, a depth-0 fact) rather than relying on complex deception chains (depth- n reasoning).

Engineering for Low Epistemic Depth. The Epistemic Saturation Proposition implies that robust systems should minimize reliance on high- n states.

- **Zero-Trust Architectures (Depth 0/1):** A Zero-Trust model assumes breach ($w_{actual} \in W_{breach}$). It does not rely on the defender knowing that the attacker knows a vulnerability ($K_D K_A V$). Instead, it enforces access controls based on immediate observables (identity, device state). Mathematically, this collapses the epistemic recursion to $n = 0$ or $n = 1$, ensuring maximal stability.
- **Deception Operations (Depth $n \geq 2$):** Conversely, honeypots and active defense strategies rely on inducing false beliefs ($K_A \neg w_{actual}$). Our model predicts that such strategies are inherently unstable (Case 1) because they rely on non-neutral objects (the deception environment) that are entangled with the observer. Therefore, deception operations must be short-lived, as the DK rate ensures the attacker’s belief state will rapidly saturate or correct itself.

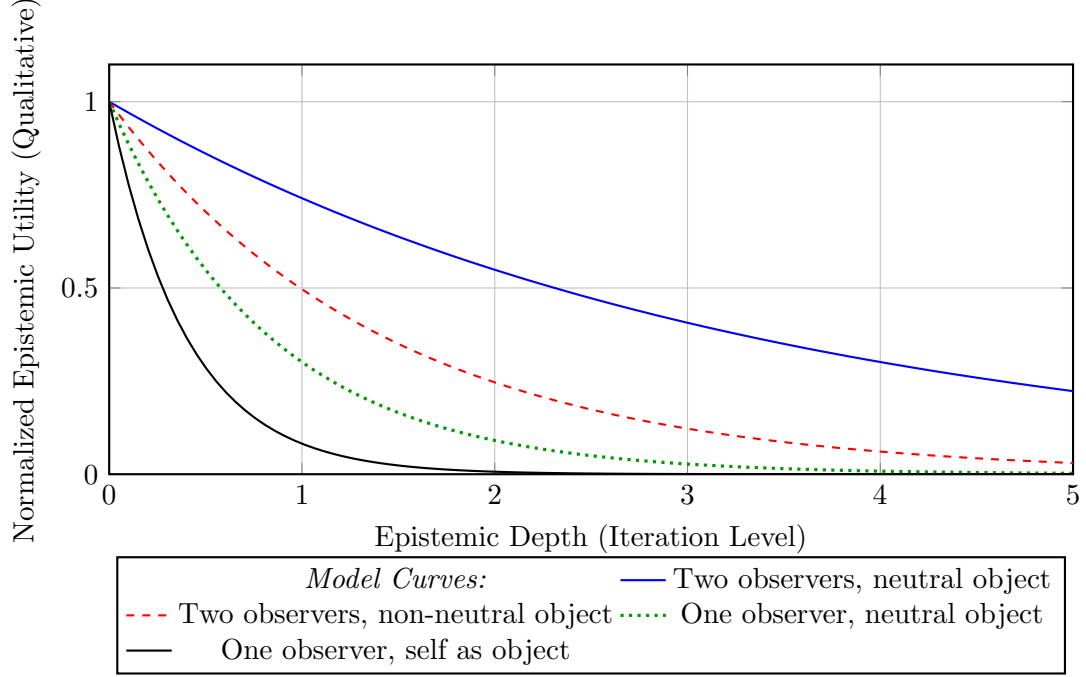


Figure 1: Qualitative illustration of epistemic decay across the four canonical configurations. The asymptote of each curve as $x \rightarrow \infty$ corresponds to the stable partition limit N derived in the **Epistemic Saturation Proposition**. Note that self-referential configurations (Case 4) reach N almost immediately, while neutral configurations (Case 2) sustain distinct partitions for longer.

8 Conclusion

This paper investigated the structural limits of iterated knowledge through the lens of category theory. By defining the category $\mathcal{E}$ as that of pointed Kripke models, we provided a rigorous basis for the intuition that knowledge decays. We proved that for finite state spaces, the filtration of epistemic states inevitably saturates, establishing the existence of a finite Epistemic Decay Bound.

This separation between formal iterability (syntax) and operational meaningfulness (semantics) is crucial for applied domains. In cybersecurity, it explains why deep recursive mind-games are theoretically possible but strategically rare. The bounds of reason are not merely psychological; they are consequences of the structural finiteness of the objects we seek to know.

References

- [1] R. Fagin, J. Y. Halpern, Y. Moses, and M. Y. Vardi, *Reasoning About Knowledge*, MIT Press, 1995.
- [2] R. J. Aumann, “Agreeing to Disagree,” *The Annals of Statistics*, vol. 4, no. 6, pp. 1236–1239, 1976.

- [3] H. van Ditmarsch, W. van der Hoek, and B. Kooi, *Dynamic Epistemic Logic*, Springer (Synthese Library), 2007.
- [4] J. A. Plaza, “Logics of Public Communications,” in *Proceedings of the 4th International Symposium on Methodologies for Intelligent Systems*, 1989.
- [5] J. van Eijck and H. van Ditmarsch, “Dynamic epistemic logic,” in *Stanford Encyclopedia of Philosophy*, 2008.
- [6] M. Burrows, M. Abadi, and R. Needham, “A Logic of Authentication,” *ACM Transactions on Computer Systems*, vol. 8, no. 1, pp. 18–40, 1990.
- [7] M. Burrows, M. Abadi, and R. M. Needham, “A Logic of Authentication,” *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences*, 426(1871), 1989.
- [8] I. Leuştean, “DELP: Dynamic Epistemic Logic for Security Protocols,” *arXiv preprint arXiv:2109.05599*, 2021.
- [9] K. Kishida, “Categories for Dynamic Epistemic Logic,” *Journal of Logic, Language and Information*, vol. 20, pp. 1–29, 2011.
- [10] G. Bakirtzis, E. Subrahmanian, C. Fleming, et al., “Compositional Thinking in Cyber-Physical Systems Theory,” *arXiv preprint arXiv:2105.12911*, 2021.
- [11] J. Vidalie, et al., “Category Theory Framework for System Engineering and Consistency,” *Applied Sciences*, 12(4), 2022.
- [12] L. A. Gordon and M. P. Loeb, “The Economics of Information Security Investment,” *ACM Transactions on Information and System Security (TISSEC)*, 5(4), 2002.

Glossary of Key Terms

Adversarial Reasoning	The process of making decisions based on beliefs about an opponent’s knowledge, intentions, or future actions. In this framework, it is modeled as iterating the knowledge functor K to anticipate the internal states of other agents (e.g., “I know that he knows that I know”).
Category Theory	A branch of mathematics that deals with abstract structures and relationships between them. In this paper, it provides the compositional language (functors, morphisms, coalgebras) to analyze how epistemic structures behave under recursive operations, independent of the specific psychological attributes of the agents.
Cybersecurity	The application domain concerned with the protection of systems from adversarial attacks. Here, it serves as the practical setting for analyzing <i>Epistemic Decay</i> , distinguishing between stable defense architectures (Zero Trust) and unstable ones (complex deception chains).
Decay of Knowledge (DK Rate)	A structural metric introduced in this paper that characterizes the loss of semantic distinction in higher-order knowledge. It measures how rapidly the sequence of partitions of the state space stabilizes. A high decay rate implies that deep iterated knowledge (K^n for large n) provides no actionable information over shallower knowledge (K^m for small m).
Epistemic Logic	A subfield of modal logic concerned with reasoning about knowledge and belief. It formalizes statements like “Agent A knows X ” using modal operators (K_AX). It provides the syntax for the constructions analyzed in this paper.
Iterated Knowledge	Knowledge about knowledge, constructed by nesting epistemic operators (e.g., $K_A K_B \varphi$). While syntactically infinite, this paper argues that operational meaningfulness of iterated knowledge is strictly bounded in finite systems.
Pointed Coalgebras	The categorical structure used to model epistemic states. A pointed coalgebra corresponds to a Pointed Kripke Model $((W, R), w_0)$, consisting of a state space W , a transition structure R (representing uncertainty/access), and a designated actual world w_0 . It serves as the fundamental object in the category $\mathcal{E}$.

A Intuitive Explanations and Analogies

Appendix A.1: The Breakdown of Meaning

This appendix explains, in concrete and intuitive terms, why deeper and deeper reasoning about “who knows what” eventually stops adding insight. The goal is not only to show that

epistemic reasoning breaks down, but to make clear *where* the breakdown occurs and *why different situations collapse at different speeds*.

Throughout, the emphasis is on structure rather than psychology. The problem is not that people become confused or inattentive. Rather, beyond a certain point, the statements themselves stop referring to anything new. They no longer distinguish between different situations in the world, even though they appear increasingly complex.

To make this visible, we examine four canonical cases by explicitly listing successive layers of knowledge and observing how meaning decays.

Case 1: Two People and a Secret (Fast Decay)

Let Tom possess a secret.

- **Layer 1:** Paul knows Tom's secret. This statement is clear and meaningful. There is an external object (the secret), and it matters who knows it.
- **Layer 2:** Tom knows that Paul knows Tom's secret. This also adds information. Tom can now reason about whether his secret has been revealed.
- **Layer 3:** Paul knows that Tom knows that Paul knows Tom's secret. At this point, the focus has shifted. Paul is no longer reasoning about the secret itself, but about Tom's awareness of Paul's knowledge.
- **Layer 4:** Tom knows that Paul knows that Tom knows that Paul knows Tom's secret. This layer adds no new actionable content. It merely loops through the same people and the same secret.

The crucial feature here is that the secret is inseparable from Tom. Every new layer pulls both agents back into the object. As a result, self-reference accumulates rapidly. After only a few layers, the statements stop distinguishing new possibilities. The reasoning becomes deeper, but not richer. This is why this configuration exhibits a fast epistemic decay rate.

Case 2: Two People and a Neutral Fact (Slowest Decay)

Now let the object be a neutral fact, such as a mathematical truth or an external event.

- **Layer 1:** Paul knows the fact. Clear and stable.
- **Layer 2:** Tom knows that Paul knows the fact. Still meaningful. The fact itself remains unchanged.
- **Layer 3:** Paul knows that Tom knows that Paul knows the fact. Understandable, though increasingly repetitive.

- **Layer 4:** Tom knows that Paul knows that Tom knows that Paul knows the fact. Interpretable, but informationally stagnant.

In this case, the object of knowledge does not belong to either person. Because the fact remains neutral at every layer, the structure is more stable under iteration. Each layer still refers back to something external.

However, even here, deeper layers eventually stop adding information. The statements become variations of the same awareness pattern. The decay is slowest in this configuration, but it is not avoided. Redundancy still appears after a finite depth.

Case 3: One Person and a Neutral Fact (Very Fast Decay)

Now consider a single person reasoning about a neutral fact.

- **Layer 1:** The person knows the fact. Clear and meaningful.
- **Layer 2:** The person knows that they know the fact. Still understandable.
- **Layer 3:** The person knows that they know that they know the fact. At this point, the object is no longer the fact. It is the person’s own knowledge state.

The critical transition occurs between the first and second layers. Once the object becomes “my knowing,” the neutrality of the fact is lost. The reasoning turns inward. Additional layers do not refine understanding of the world; they merely restate the agent’s self-awareness in increasingly indirect forms. As a result, meaning collapses very quickly—faster even than in the two-person secret case.

Case 4: One Person Knowing Themselves (Immediate Collapse)

Finally, consider the most extreme configuration.

- **Layer 1:** A person knows themselves. This is already vague. What exactly is being known?
- **Layer 2:** The person knows that they know themselves. No new clarification is added.
- **Layer 3:** The person knows that they know that they know themselves. The statement is now entirely self-referential.

There is no external anchor at any level. The structure collapses immediately. From the very first layer, the statements fail to introduce clear distinctions. Additional nesting simply amplifies repetition without adding content.

Comparative Summary

Across the four cases, epistemic decay accelerates as neutrality is lost and self-reference increases:

- Two people with a neutral object decay slowest.
- Two people with a proprietary object decay faster.
- One person with a neutral object decays very rapidly.
- One person knowing themselves collapses immediately.

These examples show that epistemic depth is not universally valuable. Its usefulness depends on how strongly the knowledge is anchored to something external.

B Why Deeper Layers Stop Helping

Appendix A.2

It is natural to assume that reasoning more deeply about who knows what should always provide additional insight. After all, understanding an opponent’s beliefs—and their beliefs about your beliefs—seems strategically important. This appendix explains why that intuition breaks down after only a small number of layers.

At shallow depths, knowledge statements are anchored to external facts. These facts divide the world into meaningfully different situations. Acting on them can change outcomes.

When a second layer is added, the object of knowledge often remains external. Knowing that someone else knows a fact can still influence decisions, such as whether to reveal information, conceal it, or act preemptively.

Beyond this point, however, a structural shift occurs. Higher layers increasingly refer not to the world, but to other knowledge statements. The object of knowledge becomes “who knows that someone knows,” rather than “what is the case.”

This shift is decisive. Once knowledge begins to refer primarily to other knowledge, iteration amplifies self-reference instead of information. Each additional layer wraps existing content in another descriptive shell without introducing new distinctions.

As a result, redundancy appears. Many different high-order statements collapse to the same practical meaning. Although the sentences look different, they do not support different actions or predictions. From a decision-making perspective, they are equivalent.

Importantly, this limitation does not arise from human weakness. Even an ideal reasoner with unlimited intelligence and computational power would encounter the same barrier. The obstacle lies in the structure of epistemic relations themselves, not in the agents who reason about them.

Another way to see this is through predictive power. Once agents have accounted for the relevant external facts and the immediate awareness of other agents, further nesting does not constrain behavior any further. The deeper layers describe beliefs about beliefs, but these descriptions do not narrow down what will actually happen.

For this reason, epistemic depth saturates quickly. There exists a maximal useful depth beyond which further reasoning does not improve understanding, prediction, or strategy. Beyond that point, epistemic reasoning becomes inert.

Summary. Deeper epistemic layers stop helping because they recycle existing information rather than introducing new, externally grounded distinctions. Epistemic depth is therefore not an unlimited resource, but a structurally bounded one.